

MEDIKIOSK

Project Handbook

Key product, clinical and architecture facts

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Use this shorter guide for demos, revision, team onboarding and judge questions. It contains the important facts without the full study detail.

Problem	Solution	Boundary
OPD time is consumed by basic history and paper records	Collect and structure information before consultation	AI drafts; clinician verifies, signs and decides

01 The project in one page

10 seconds

MediKiosk is a patient-facing pre-consultation kiosk that gathers structured history, digitizes paper records, flags urgent patterns and prepares a clinician-reviewable summary.

30 seconds

Patients use English or Hindi in Speak or Chat mode. A server-controlled interview follows SOCRATES and Ayurveda-specific fields while a local LLM phrases questions and extracts answers. Deterministic safety rules can alert the Nurse Station. The doctor reviews, corrects, signs and only then creates a PDF or FHIR export.

Architecture in one line

React/Vite -> FastAPI:8080 -> SQLite + Ollama llama3.1:8b + faster-whisper/Piper + EasyOCR -> clinician sign-off
-> ReportLab PDF / FHIR R4 document

Maturity

Strong functional hackathon prototype with production-minded controls. It is not a deployed clinical system, certified ABDM connection, diagnostic device or autonomous doctor.

02 Problem statement coverage

Status	Covered areas
Covered	Consent-first flow; adaptive voice/touch; SOCRATES; structured summary; clinician edits/revisions/sign-off; red flags; audit; FHIR document generation.
Partial	English/Hindi only; Ayurveda strong but other AYUSH profiles and full physical-exam workflow pending; OCR handwriting always reviewed; abnormal lab and chronology limited; consent revocation and bilingual clinician output incomplete; ABDM/HIS staged.
Missing	Drug-interaction engine; Aadhaar flow; production HIS connector; certified live ABDM exchange.

Answer to judges

We cover the complete prototype journey and all four major modules, with strong clinician governance. We do not claim literal production completion: drug interactions, Aadhaar, broader languages, granular revocation, full Ayurveda examination workflows and live integrations remain planned.

03 End-to-end workflow

Stage	Result
Start -> consent	Bilingual explanation; accepted scopes and timestamp create the session.
Language -> mode	English/Hindi and Speak/Chat; orb centers only in Speak.
Identity	New Medi ID or returning lookup; optional ABHA; sensitive numbers use touch.
Department	General or Ayurveda department selects the clinical profile.
Interview	Automatic prompt/listen/confirm in Speak; typed touch flow in Chat; captions in speech.
Safety	Deterministic/AI checks can create a live Nurse Station alert.
Read-back	Patient confirms a short spoken summary or dispute alerts staff.
Documents	OCR extracts reviewable diagnoses, medicines and lab values from images.
Clinician	Review, correction reason, revision, Ayurveda confirmation, attestation and sign-off.
Output	Patient/clinician PDF or validated FHIR candidate; kiosk credential invalidated after export.

04 Clinical and Ayurveda essentials

SOCRATES

Site, Onset, Character, Radiation, Associated symptoms, Timing, Exacerbating/relieving factors and Severity. Medicines and allergies are deliberately collected early, followed by past medical/surgical, family, Ahara-Vihara/personal and systems history.

Ayurveda

Provenance	Fields
Patient-reported	Vikriti, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya; plus Agni, Koshta, Nidana and Panchakarma history.
Practitioner-confirmed	Prakriti.
Practitioner examination	Sara, Samhanana, Pramana.
Protected placeholders	Trividha: Darshana, Sparshana, Prashna. Ashtavidha: Nadi, Mutra, Mala, Jihva, Shabda, Sparsha, Drik, Akriti.

The product currently supports Ayurveda departments, not every AYUSH discipline. It never asks the LLM to invent examination findings.

05 Architecture and safety decisions

Owner	Responsibilities
Frontend	Touch UI, voice/orb state, captions, accessibility, dashboard and device/fleet views.
FastAPI	Identity, cookies, roles, field schedule, validation, merging, alerts, persistence, sign-off and export.
LLM	Natural question phrasing and constrained extraction only.
Speech	faster-whisper ASR and Piper TTS; Bhashini/AI4Bharat adapters with local fallback.
OCR	EasyOCR plus constrained entity extraction and mandatory uncertainty review.
SQLite	Prototype authority for patients, visits, evidence, staff, alerts, audit, revisions and integration state.

Safety spine

- Server chooses coverage; model cannot skip requirements.
- Deterministic red flags survive model failure or prompt injection.
- Patient/OCR text is treated as untrusted data.
- Unknown data stays unknown; handwriting needs staff review.
- Every AI summary remains a draft until current clinician sign-off.
- Later edits invalidate old sign-off and export.

06 ABDM, FHIR and consent

Term	Remember this
Medi ID	Local patient identifier; works without ABHA.
ABHA	Optional national health identity link; real OTP/QR is pending.
ABDM	Consent-based national exchange ecosystem; local/staged today.
FHIR R4	Exchange format. MediKiosk creates a document Bundle with an OPConsultRecord Composition first.
HIS/EMR	Hospital system that should receive the approved record; real connector pending.

Production ABDM still needs registered facility/bridge context, issued credentials and paths, real ABHA verification, consent-artifact cryptography, official validation, certification and secure inbound provenance review.

Consent today

Bilingual audio/text notice, stored scopes/time, HttpOnly cookie, idle/manual clear, post-export credential invalidation, registry retention/purge and erasure endpoints. Granular per-scope withdrawal and a visible consent wallet are not built.

07 Security, audit and operations

- Nurse, Doctor and Admin roles with server-side authorization.
- Hashed patient/staff tokens; HttpOnly SameSite=Strict patient cookie; Secure when TLS is configured.
- Masked identifiers, strict schemas, upload limits and OCR pixel cap.
- Rate limits for staff login, Medi ID lookup and Help.
- Temporary voice recordings deleted after transcription.
- Append-only audit events linked by previous hash and SHA-256 current hash.
- Kiosk IDs, heartbeats, fleet status, alert escalation and device diagnostics.
- Graceful backend/Ollama failure screens and full-screen kiosk launcher.

Before deployment

Replace SQLite, require TLS, encrypt data at rest, add secrets management, backups/recovery, monitoring, high availability or degraded mode, external staff paging, formal retention policy, physical commissioning, clinical validation and legal/security review.

08 What works, what remains

Strong now	Next critical work
Consent-to-sign-off patient and clinician workflow	Clinical/usability study with real OPD users and clinicians
Deterministic clinical orchestration + bounded LLM	Drug-interaction and deterministic lab-safety engine
Two-layer red flags + Nurse Station	Validated safety dataset and external escalation
Ayurveda provenance model	Full Trividha/Ashtavidha workflow and separate AYUSH profiles
OCR review states	Hospital document benchmarks and dedicated procedure extraction
FHIR document and ABDM boundary	Live HIS/ABDM, cryptography and certification
English/Hindi voice/chat	Regional languages and streaming conversational voice
Security/audit foundation	Production infrastructure, governance and operations

Explicit missing requirements

- Aadhaar identification.
- Drug interactions.
- Full bilingual clinician output.
- Per-scope consent revocation.
- Production HIS and ABDM exchange.

09 Terms to know

Term	Meaning
OPD	Outpatient Department.
SOCRATES	Eight-part symptom-history framework.
LLM	Large language model.
ASR/STT/TTS	Speech recognition / speech-to-text / text-to-speech.
OCR	Text recognition from document images.
RBAC	Role-based access control.
Provenance	Source, actor, time and verification state of a fact.
ABHA	National digital-health identity/account.
ABDM	India's digital-health ecosystem.
FHIR R4	Healthcare exchange standard/resource model.
HIP/HIU	Provider/user roles in ABDM exchange.
HFR/HPR	Facility/professional registries.
SNOMED CT	Clinical terminology.
LOINC	Lab/observation terminology.
DPDP	India's personal-data protection law/rules.
Dashavidha	Ten-factor Ayurveda assessment.

10 Demo and viva cheat sheet

Recommended demo

- Run Device Check and health status.
- Start, hear bilingual consent, choose Hindi/English and Speak, show orb/captions/automatic listening.
- Register or retrieve a patient, choose an Ayurveda department and answer a short SOCRATES/Ayurveda path.
- Show a printed scan and the mandatory handwriting-review route.
- Trigger a prepared red flag and acknowledge it at Nurse Station.
- Open Physician View, correct a field with a reason, show revision, confirm Ayurveda fields and sign.
- Generate PDFs and show the local validated FHIR Bundle without claiming live delivery.

Five answers to memorize

- AI helps conversation and extraction; application code owns safety and workflow.
- The product prepares history; it does not diagnose or prescribe.
- ABHA is optional identity; Medi ID keeps the local workflow usable.
- FHIR is the exchange format; ABDM/HIS are destinations and workflows.
- Core PS prototype coverage is strong, but named production and clinical gaps remain.

Current evidence

Latest recorded verification: 43 backend regressions, 17/17 clinical safety cases, production frontend build, 6/6 contrast checks and a live light/dark/touch/voice/viewport walkthrough.

Reference: the Complete Guide V4 contains the full coverage matrix, operating instructions, study map, risks and sources.

11 Reference links

Reference	Location
MediKiosk current scope	ROADMAP.md and docs/ in the repository
Clinical schema	docs/schema.json
ABDM FHIR profiles	https://nrces.in/ndhm/fhir/r4/
HL7 FHIR R4 Bundle	https://hl7.org/fhir/R4/bundle.html
ABDM	https://abdm.gov.in/
MeitY DPDP material	https://www.meity.gov.in/

Final mental model

Patient gives consent, history and documents -> MediKiosk structures them -> deterministic safety watches continuously -> clinician verifies and signs -> approved data becomes a PDF or FHIR document -> HIS/ABDM is the integration destination.

End of handbook